

## Being Fruitful Tenants

October 8, 2023, Thanksgiving

Matthew 21: 33-46

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In 2019, the Canadian government purchased the Trans Mountain pipeline for \$4.5 billion from Kinder Morgan Canada Inc after the company threatened to scrap the pipeline expansion project due to environmentalist and indigenous opposition and political uncertainty. This was done at a time when we are supposed to be looking for ways to reduce our production and consumption of fossil fuels to reduce our emissions to keep the global temperature below a 1.5°C increase from industrial levels. Since this purchase, the cost of the project to the government has soared from 21.4 billion in early 2022 to 30.9 billion in March this year. It seems to some that this was an irresponsible decision by the government, the powers that be, whom we have entrusted to manage our country and its' resources. Is this really going to bear fruit for the country, for the world? Jesus might have a similar challenge to our government about this as he did to the religious authorities of his time.

In our gospel reading today, Jesus tells a parable about tenants of a vineyard who, instead of passing the harvest over to the landowner, killed the servants and then his son, to try and secure the harvest for themselves. Jesus asks the chief priests and scribes listening, what should that landowner do with the tenants? They respond that they should be killed and the vineyard given to tenants who will be more responsible and give the harvest to the landowner. Jesus then challenges them that the kingdom will be taken away from them and given to those who can be more responsible in producing fruits of the kingdom. The religious authorities realize Jesus is talking about them, that they are the tenants in the story, and want to arrest him but cannot because of the crowds. Jesus' challenge to them in this parable and the previous one about two sons, each who do the opposite of what they said they would do when asked, indicates that they are not living out God's will. We know from Jesus' previous teachings and actions that he understands God's will to include being loving and compassionate to those in

need, ensuring that everyone is cared for and has enough, that they are paid enough to feed their families, that there should not be any corruption or extortion, especially in the temple. If they were following God's ways, the community would be bearing fruit.

The wider message here is that it is our responsibility to care for what God has entrusted to us – one another and all of creation. Carrying out this responsibility should bear fruits of the kingdom – justice and equity, love and compassion. As we work toward this goal and calling, there is much in our lives and in creation we can be thankful for. Caring for all that God has entrusted to us, and being responsible also includes living with gratitude for what we have. As we celebrate thanksgiving this weekend, it is good to pause and consider all that we are thankful for, and sit with this gratitude for a moment.

We are thankful for a loving and compassionate community of faith.

We are thankful for families, birth and chosen, who love and support us.

We are thankful for friends who are there when we need them, whether for support or fun.

We are thankful for food and shelter, and those who care for those without.

We are thankful for faith and hope, and for those close to us who have it when we don't.

We are thankful for opportunities to gather and learn and grow together.

We are thankful for generous hearts and donations that meet needs within our community and beyond.

We are thankful for the beauty of God's creation and all the ways we can enjoy it.

We are thankful for life in all its fullness and the blessings we experience each day.

The tenants in the story seemed to have missed this kind of gratitude. They were so desperate for what they wanted, the harvest, that they took action that did not make sense. How would killing the son of the landowner, get them his inheritance? Especially when the landowner is still alive. Sometimes we get so focused on something we want, or think we want or need, that we behave in ways that seem counterproductive or do things that appear illogical to many. Can you think of examples of this in your life? In our society? Here are some of my musings about this:

- Could the government buying a pipeline and continuing to consume fossil fuels at the same rate, knowing the disasters the greenhouse gas emissions are causing be seen as counterproductive?
  - How about using the notwithstanding clause of the Charter of Rights to limit the human rights of others?
  - Maybe the banning books and drag queens from reading books to children to avoid the so called sexualization of children.
  - Closer to home, what about tearing down tents where the unhoused were sleeping, and not providing any other options to address the lack of shelter.
- The Canadian Government is now looking to sell the Trans Mountain pipeline to Indigenous communities – some of whom would buy because they need the economic opportunity even though many other Indigenous communities are against it and it goes against their values of caring for the land. Here, I would say that the Government is taking advantage of the vulnerability of indigenous communities who need economic opportunities, instead of addressing the roots of their poverty and finding solutions more in line with their culture and traditions. We have much work to do to bear the fruit of the kingdom in our lives and leadership.

God calls us to love and compassion which will bear fruit of the shalom community, where all have enough, where justice and righteousness is a priority; where people are housed and cared for and get the health care they need. May we be up to the challenge of carrying out this call and calling out the places, people, and organizations which are falling short of God's will. May we live with gratitude for all we have and may that gratitude motivate us to care for one another and all of God's creation.

# Implications of LEP and LEPR Within the Human Body Pertaining to Obesity

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## Key Words:

Leptin, leptin receptor, and obesity.

## Abstract:

The critical examination of the following study by Fairbrother aims to review the current information on genes and their expression regarding their involvement with obesity. With a specific focus on the genes LEP and LEPR, an underlying basis of contextualization, interpretation, and totality are implemented to further the comprehension on just how essential these genes are as a causative agent of glucose levels and metabolism regulation, thus impacting body weight. Variants in genes are present and their involvement in signaling pathways present that monogenic variants are contributive in obesity. Identifying such genes that are responsible for severe obesity and the distinction between types of obesity is now conflated as rare and common obesity share similarities of variants across the general population. This is not unforeseen as leptin is a complex protein; such technological advancements in genotype and their sequencing allows them to be associated with a character for all variants.

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## Introduction:

—Obesity has seemed to situate itself in a customary notion in modern time becoming more pervasive. Obesity is concomitant with an ample number of complications such as cardiovascular disease, type 2 diabetes, and cancer. Although its underlying causes have been difficult to elucidate, it is evident that obesity and one of its causative agents is influenced by such genetic components. The human monolog of the gene is characterized by a capitalized O. Encoded by the LEP gene and through its receptor LEPR, both genes are instrumental in the transduction of leptin messages via signaling pathways. As a complex phenotype (derived from more than one gene) in which behavioral, environmental, and genetic features all play a role, leptin is a 167 amino acid that is an enzyme; it suppresses appetite and galvanizes metabolism. It is essentially a peptide that is released by fat cells. Other functions and capabilities of leptin include its role of acting as a hormone requiring a specific cell receptor for binding before they cause the effect. Leptin is a cascade of events entailing the process of leptin stimulating the hypothalamus, the hypothalamus embracing the pituitary gland, the pituitary gland invigorating the thyroid gland to

release thyroxine, thus raising the metabolism rate. However, there are several factors which inhibit this enzyme from operating optimally and correctly. Induction of a cascade to generate the thyroid gland affects NPY suppression of appetite (Marley, 2018); the suppression of neuropeptide y (NPY)—an appetite stimulant—results in the loss of appetite. Moreover, leptin blocks transcription of mRNA for NPY, resulting in appetite decreasing. In addition, leptin has the possibility of making the intramitochondrial membrane pregnable or leaky to protons. By stimulating the release of uncoupling proteins, the membrane is therefore weakened. Furthermore, this leads to the rapid release of electrons from the body as thermogenesis; the lack of ATP increases the demand for more electrons from carbohydrates and fats, causing more carbs and fats to enter the respiratory pathway. Scotee states that leptin also conditions inflammation, immune responses, infection, and bone metabolism; all which are physiological features (Scotece et al., 2014). Correspondently, Procaccini mentions the LEPR gene being expressed across somatic cells associated with the innate and adaptive immune system, eliciting the gene leptin as an imperative bridge of immune systems and neuroendocrine (Procaccini et al., 2017). The scientific report by Fairbrother that I intend on analyzing encompasses the focus of obesity as an idiosyncratic disease with a specific genetic cause rather than common polygenic obesity.

—Recent studies from as early as this year by Ali and company (Ali et al., 2019) address the role of fat mass, leptin receptor, and obesity concomitant polymorphisms for the impressionability of obesity among various individuals; by using the methodology of PCR genomic DNA was genotyped to illustrate the relationship of the gene and obesity.

## **Discussion:**

—Upon reviewing, Fairbrother establishes the connection between severe obesity and postulates theory of a specific disease in conjunction with a particular gene(s). LEP and LEPR are illuminated on as variants that give rise to obesity and monogenic obesity within the universal population. In a study by Adams, selected genes with variants showed to contribute to acute obesity (Adams et al., 1993). Tests with mouse at the helm as subjects implicated leptin and its receptor in monogenic obesity. Fairbrother and company also stress the hormone cascade in which leptin acts and influences the transcription factors in relation to homeostatic pathways, including sugar and calorie levels. In addition to the behavioral portioning regarding food intake, leptin induces its effects of another receptor, LepRb. LepRb is specifically expressed in the region of the brain responsible for energy expenditure and food regulation. The complexity of leptin is extrapolated in this report. Fairbrother and all are not linear in approaching the signaling process. While the rudimentary disposition focuses on leptin pertaining to obesity or individuals with a significantly above average weight, individuals with a normal weight or body mass are not just used as the control experiment for comparative measures. Normal body weight people are mentioned with fluctuating levels of leptin; reported is decreased levels of leptin lead to a direct relationship of decreased levels of sugar and energy use. This highlights the functionality of leptin and leptin receptors as a main component in metabolism and glycaemic control.

Shifting its focus back on LEP and LEPR in obese individuals, Fairbrother involves pathways mechanism factors that have multitude levels of influences such as inflammatory responses and

feedback inhibitions. Leptin levels are increased in obese humans and animal models, but the imperative feedback loop that generates reduction in the behavior of food and great energy expenditure fails. Interestingly, Fairbrother comes to the revelation that an obese final point is reached despite increased leptin to leptin receptor gesturing, and that increased levels of leptin signaling eventually creates leptin resistance limiting the full scope of the amplitude of the process. This is mentioned by Al-Suhaimi and Shehzad as they report how leptin resistance caused by impairment of leptin signaling and transportation is thought to be the central risk factor for the obesity pathogenesis (Al-Suhaimi and Shehzad, 2013). Allelic variation and mutation are even discussed examining the genetic impact on identification of positive associations of leptin in association with body maintenance.

Furthermore, such genes, like BMI and MC4R, and their expression have been found to be in people who are similar in phenotype (obese). This study attests to the newly found comprehension and relatedness between obesity and genetics and implicates for the future variants within the genome that can be worked upon in a nuanced fashion broaching the issues obesity proposes on the population.

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